9 Implementation of Subnetting in CISCO Packet Tracer Simulator.

Aim: implementation of Subnetting in CISCO Packet tracer Simulator

Class less IP subnetting is a technique that allows for more efficient use of IP address by allowing for subnet masks that are limited number of IP addresss but need to create multiple networks.

Creating a network topology:

The first Step in implementing class IP Subnetting is to create a network topology in Packet Tracer. To create a network topology in Packet Tracer. Select the New button in the top left corner

ADDing the Devices:

Once we have created our network topology We can add devices to it. Here we will be adding routers, witches and PCs. To add a device. select the device from the bottom left corner and drag it onto the network topology. Then connect the devices by drawing a cable from one device's port to another device's port.

Subnetting :

To Subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for 4 end device the Switch and the router, we can use a /2 Subnet mask. This will give us 8 subnet with 30 host address each.

The addressing for the network shown in the topology can be as follow.

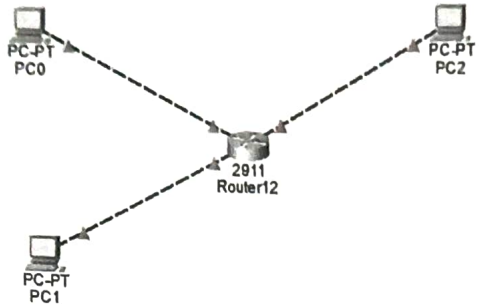
- Router R1:
- Gigabit 0/0 : 192.168.1.1
- Gigabit 0/1 : 192.168.2.1
- Switch S1:
- fast Ethernet 0/1 : 192.168.1.0/27
- PC1 : 192.168.1.17
- PC2 : 192.168.1.12
- PC3 : 192.168.1.13
- PC4 : 192.168.1.14
- PC5 : 192.168.1.15

Fast Ethernet 0/2 : 192.168.2.0/27

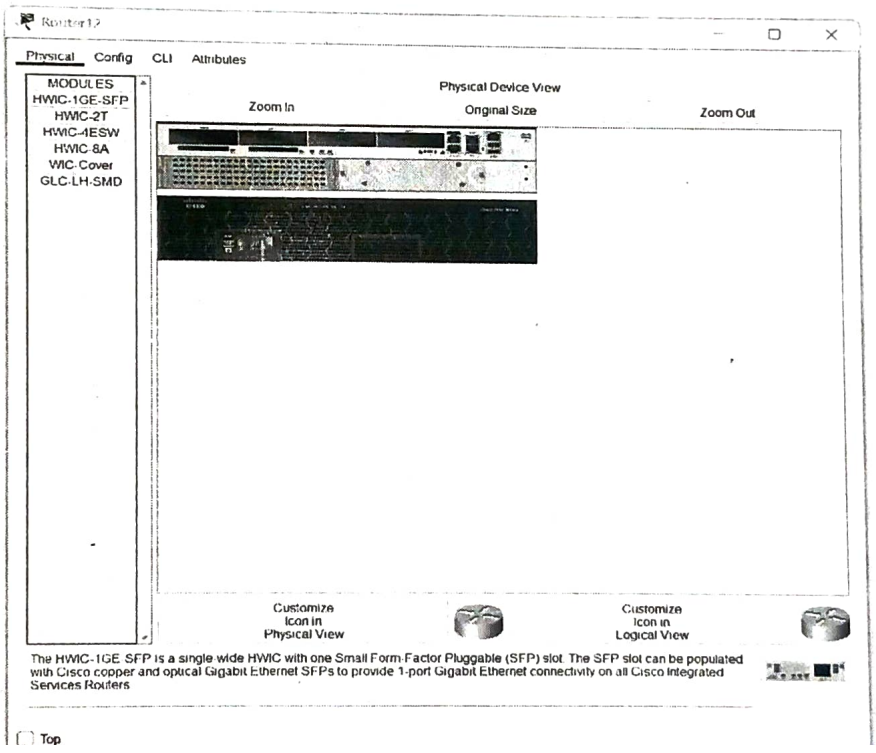
- PC1 : 192.168.2.11
- PC2 : 192.168.2.12
- PC3 : 192.168.2.13
- PC4 : 192.168.2.14
- PC5 : 192.168.2.15
- Router R2

Fast Ethernet 0/0 : 192.168.3.1

Fast Ethernet 0/1 : 192.168.4.1



From this Experiment 9



```
Router>running-config
Translating "running-config"...domain server (255.255.255.255) % Name lookup aborted
Router>en
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#show running-conf
Building configuration...

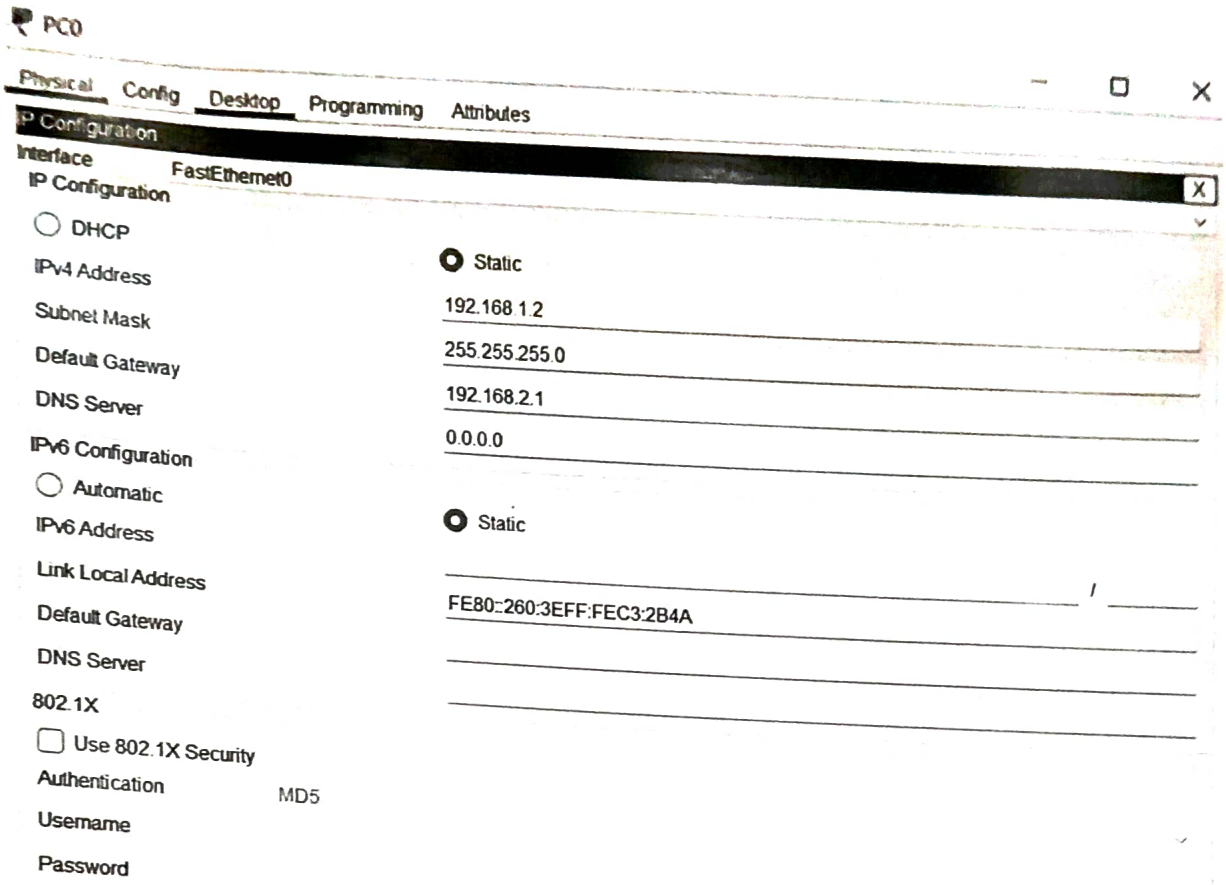
Current configuration : 732 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524WFBP-
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.2.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 ip address 192.168.3.1 255.255.255.0
 duplex auto
 speed auto
!
interface Vlan1
```

```
Router#show startup-conf
Using 732 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524WFBP-
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.2.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 ip address 192.168.3.1 255.255.255.0
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!

Router#
```

```
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#show running-conf
Building configuration...

Current configuration : 732 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:3EFF:FEC3:2B4A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

configuring the Device:

enable
configure terminal
interface FastEthernet 0/1
ip address [IP address] [subnet mask]
no shutdown
exit.

Replace {IP address} and {subnet mask}
with your desired IP address and Subnet mask.
The first interes with IP address and Subnet mask

Testing the Network

Now that our network topology is configured
We can test the network. Open a command prompt
on each PC and try to ping the other PC. If Ping
is success full, then the network is functioned
Properly. We can also use the ping command
to test connectivity between the router
and PCs.

Result: Implementation of Subnetting has
been done Successfully using cisco packet tracker